Group Members:

STUDENT NUMBER	INITIALS & SURNAME
224067603	JO Molaba
224093660	MN Dlamini
224081110	M Lesenyeho
224022386	LW Simelane
222021446	PA Mabusela
221027626	M Mokhele
223000608	AL Hadebe
223014209	L Sandi
218013010	NP Khotle
222044762	NL Mazibuko

1.1 Purpose of the Project

Big Brother Student Accommodations currently experiences difficulties in advertising accommodation, sharing accommodation information, managing student applications, and maintaining accommodation records. Most processes are handled manually and rely heavily on word-of-mouth communication, WhatsApp groups, and referrals. The purpose of this project is to develop an integrated accommodation management solution that improves access to accommodation information, simplifies applications and bookings, and assists management in monitoring accommodation operations.

1.2 Proposed Solution

The proposed solution consists of:

- Flutter Mobile Application
- ASP.NET Web Application
- Shared Supabase Backend

The two applications will share information through a centralized Supabase database, ensuring data consistency between students and accommodation management.

STAKEHOLDER PROFILE AND CURRENT PROCESS

ORGANISATION INFORMATION

Detail	**Information**
Business Name	Big Brother Student Accommodation
Location	Universitas, 50 Besselsen St
Stakeholder	Mr Ngubane (Owner)
Contact	068 518 4009

STAKEHOLDER PROFILE

Name: Mr Ngubane

Role: Owner/Property Manager

Responsibility: Full Operational management, student sourcing, bookings, payments, data management

Current Admin Time: Manual Systems with no proper management tools in place

USERS

User Type	Description	Responsibilities
Primary User (Management)	Mr Ngubane (Owner/Manager)	Control, monitor, manage data; access student information; analyse and respond to applications; add and update accommodation information
Secondary Users (Students)	Prospective and current students	Registers, view accommodations, check availability, apply, make bookings, view status updates, upload particulars

Existing Process:

1. **Enquiry and Viewing**
    - Students are encouraged to physically visit the accommodation for viewings
    - All facilities and services offered by the accommodation are explained to students during the visit
    - Students decide whether they are satisfied with the accommodation based on the physical viewing
    - No digital platform for virtual viewings or tours
    - No follow-up system is in place for students who interest
2. **Advertising and Marketing Process:**
    - Advertising is currently done primarily through word of mouth
    - Student WhatsApp groups are used to share accommodation information
    - Taxi association organisers are utilised to advertise to students travelling from home to university
    - Current students are requested to advertise the accommodation to prospective tenants
    - Group admins are used to advertise the business through various platforms

3. **Booking Management Process:**
   - Bookings are managed manually with no digital system to handle accommodation requests
   - There is no centralised record of all bookings, making it difficult for the owner to track who has booked and which units are occupied.

4. **Payment Management Process:**

   - Payments are managed through a manual system
   - No proper digital payment tracking system exists
   - Payment records are not centralised or easily accessible

5. Student Application Process:

   - Student are required to submit their personal information and documents manually, usually by sending scanned copies via WhatsApp or email
   - There is no digital application system in place where students can simply login, fill in their details, and apply for accommodation

6.Maintenance Management:

   - Students submit requests via WhatsApp or text messages
   - Requests are tracked in memory or through WhatsApp history
   - No system to track status (open, in-progress, completed)
   - Requests often get forgotten or overlooked

7.Occupancy Tracking:

   - Relies on paper records that are frequently out of date
   - No quick reference system for identifying current occupants
   - New tenants and guests are difficult to track

Problem Statement

Current Situation:

Big Brother Student Accommodation is a student housing business located in the Universitas area of Bloemfontein, owned by Mr Ngubane. The business currently operates using manual processes, word-of-mouth advertising, and informal communication channels. The business relies on physical viewings for students to see the accommodation, and advertising is limited to WhatsApp groups, taxi associations, shuttle-group administrators, and referrals from current students.

There is no digital system in place for students to view accommodations online, submit applications, make bookings, or track payments. Bookings and payments are managed manually, with no centralised platform to store or manage student information. Management currently uses WhatsApp, email, and physical records to handle all operations, resulting in scattered and disorganised data.

The business also faces significant challenges in sourcing students, as the current advertising methods do not reach the intended number of prospective tenants. First-year students are particularly vulnerable to scammers who take advantage of the absence of an official application platform. The business is struggling to remain sustainable, and the stakeholder has indicated that these challenges pose a real threat to the continuity of the business.

Problem:

Big Brother Student Accommodation lacks an integrated digital system to manage its core operations, resulting in the following specific problems:

Identified Problems:

- **Difficulty sourcing students** - The business struggles to reach the intended number of students. Currently, advertising is mostly done through word of mouth, WhatsApp groups, and taxi associations. This makes it difficult to attract enough students, especially those who are new to the university or coming from other areas. As a result, rooms often remain empty, which affects the business financially. Many students simply do not know that this accommodation exists.

- **Inaccessibility for Students** - Students cannot easily view the accommodation before deciding. The business relies on physical viewings, which are not always possible for students who are far away or do not have transport. This discourages many students from considering the accommodation because they cannot see what they are getting. It also creates a barrier for first-year students who may not yet be familiar with the area.

- **First-year students are vulnerable to Scammers -** Because there is no official online platform for accommodation applications, first-year students are often targeted by scammers who pretend to help them find accommodation. These scammers take money from desperate students and disappear, leaving them without a place to stay. This damages the reputation of legitimate accommodation providers and makes students afraid to trust anyone when looking for housing.

- **Difficulty sharing information -**Students cannot easily find information about the accommodation online

- **No proper management system -** Bookings and payments are manual, causing confusion and delays

- **Business continuity threat -** These challenges pose a real threat to the existence of the business. If students cannot find the accommodation, cannot easily apply, or do not feel safe using the current process, they will look elsewhere. The business may struggle to remain sustainable if it cannot attract and retain enough students each year**.**

- **No application platform -** Students cannot apply online or track their application status

- **Limited access to information** - Management cannot easily access or respond to student applications

Affected Users:

User	Impact
Students	Cannot easily find information, view accommodations remotely, or apply securely. Vulnerable to scammers. No way to track application status.
Management	Manual administration, disorganised data, no centralised platform, inefficient operations, business sustainability at risk.
First-year Students	Particularly vulnerable to scammers, at risk of financial loss and housing insecurity.
Business	Unsustainable operations, risk of closure, reduced accommodation options for students.

Consequences:

The current manual and fragmented systems at Big Brother Student Accommodation have resulted in several serious consequences that affect both the business and its students.

Firstly, the business struggles to source enough students due to ineffective manual advertising methods such as word of mouth and WhatsApp groups, resulting in empty rooms and lost revenue. Secondly, students, particularly first-years, are vulnerable to scammers who exploit the lack of an official application platform, damaging the business's reputation and causing financial loss and distress for students. Thirdly, the business cannot easily share important information about available rooms, prices, and facilities with prospective tenants, reducing its visibility and reach, and causing potential tenants to choose other providers.

Fourthly, the absence of a proper management system places a significant administrative burden on the owner, who must manage bookings, payments, and applications manually, leading to errors and inefficiency. Fifthly, these cumulative challenges threaten the existence of the business, making it unsustainable and at risk of closure. Finally, the lack of an application system results in inefficient application processing, leaving students frustrated and uncertain about their applications. These consequences highlight the urgent need for an integrated digital solution to address these challenges and ensure the sustainability of the business.

Need for Improvement:

A unified digital system is required to address these challenges. The proposed system will enable students to upload their particulars and apply for accommodation through a centralised platform. Management will be able to control, monitor, and manage data effectively. The system will allow prospective tenants and management to access and view the entire application process, business model, and operations. This will reduce vulnerability to scammers, improve student sourcing, and ensure business continuity.

Project Aim and Objectives

Overall, Purpose: To develop an integrated software prototype that digitises and streamlines the management operations of Big Brother Student Accommodation, improving student sourcing, application processing, and operational efficiency.

Specific Objectives:

Objective	Action
1. Digital application platform	Enable students to upload particulars and apply through the system
2. Accommodation viewing	Allow students to view accommodations, check availability, and make bookings
3. Centralised management	Enable management to control, monitor, and manage data
4. Information sharing	Allow easy sharing of business information to prospective tenants
5. Security	Secure student information and ensure it is not accessible by unauthorised persons
6. Application tracking	Allow students to check status updates accordingly
7. Data analysis	Enable management to access, analyse, and respond to student information
8. Student self-service	Allow students to view, add, and change their information
9. Business visibility	Allow students to understand vision, values, and mission of the organisation
10. Payment system	Implement payment functionality
11. Feedback mechanism	Collect reviews, comments, and areas of improvement from students
12. Unit management	Track units available, furniture, occupant details, deposits, and breakages
13. Maintenance management	Allow students to submit maintenance requests and management to track them
14. Announcement system	Allow management to post announcements and students to receive and view notifications
15. Viewing request management	Allow management to manage physical viewing requests from students

Project Scope

Deliverables:

- Flutter mobile application for students
- ASP.NET Public web application for management, visitors and students
- Supabase shared backend and database
- Integrated solution where both applications access the same data
- Technical documentation
- User guides where applicable

Acceptance Criteria:

- Management can view and manage student applications
- Management can control, monitor, and manage data
- Management can access, analyse, and respond to student information
- Management can view accommodation listings and occupant details
- Visitors can view accommodation information and room details without creating an account.
- Students can upload particulars and apply on the system
- Students can view accommodations, check availability, and make bookings
- Students can view, add, and change their information (with management approval)
- Students can submit reviews, comments, and recommendations
- Students can submit maintenance requests
- Students can view notifications
- Students can upload necessary supporting documents for their accommodation applications
- Student information is secured and not accessible by unauthorised persons
- All information is consistent across mobile and web applications
- System is functional for demonstration purposes

Exclusions:

- Full payment processing integration (e.g., automated EFT reconciliation)
- Advanced reporting and analytics dashboards
- Integration with external systems (e.g., banking APIs, municipal systems)
- Full production deployment and hosting
- Professional security audit and penetration testing
- Mobile app store publication
- Multi-property support (limited to Sunrise Student Accommodation)

Project Constraints:

- **Time:** Development must be completed within one academic semester
- **Budget:** Limited budget (student project)
- **Technology:** Stack is predetermined: Flutter, ASP.NET, Supabase
- **Skills:** Team comprises of IT students possessing Flutter, Dart, ASP.NET, Github and Supabase skills necessary for development of the system.

Project Assumptions:

- Stakeholder will be available for requirements gathering, feedback, and evaluation
- Student users will be accessible for feedback and testing
- Supabase will provide sufficient functionality for the project scope
- The approved prototype does not need to be production-ready
- All development work will be managed through GitHub
- Microsoft Project will be used for project scheduling

Stakeholder Needs:

- Students should be able to upload their particulars and apply on the system
- Students should be able to view accommodations, check availability, apply, and make bookings
- Students should be able to check status updates
- Students should be able to submit maintenance requests
- Students should be able to view notifications and announcements
- Students should be able to upload the necessary supporting documents for their accommodation applications.
- Management should be able to control, monitor, and manage data
- Management should be able to access, analyse, and respond to information
- Management should be able to manage viewing requests
- Management should be able to manage rooms, units, and maintenance
- Management should be able to post announcements
- Prospective tenants should be able to use the application
- Management should be able to use the application
- Students should be able to view the entire processes and business model
- Students should understand the vision, values, and mission of the organisation
- Student information should be secured and not accessible by unauthorised persons
- Easy access to information and application for accommodation
- Payment system should be included
- Students should be able to view, add, and change their information (with management approval)
- Management should be able to view accommodation listings and occupant details
- System should provide summary of students visiting page and viewing accommodation
- System should collect reviews, comments, and areas of improvement from students
- System should track units available, furniture, occupant details, deposits, and breakages

Functional Requirements:

- System shall allow students to create accounts and upload their particulars
- System shall allow students to log in and authenticate their identities.
- System shall allow students to submit accommodation applications
- System shall allow students to view available accommodation listings.
- System shall allow students to check accommodation availability and make bookings.
- System shall allow students to check application status and view status updates.
- System shall notify students of important updates.
- System shall allow students to view notifications
- System shall allow students to submit maintenance requests
- System shall allow students to submit reviews, comments, and recommendations for improvements
- System shall allow management to control, monitor, and manage data.
- System shall allow management to access, analyse, and respond to student information.
- System shall display the entire application process and business model.
- System shall display the organisation's vision, values, and mission.
- System shall authenticate student and management users and enforce access permissions according to their assigned roles
- System shall validate user input before accepting or processing submitted information.
- System shall display appropriate error messages when invalid information is entered or an operation fails.
- System shall secure student information
- System shall provide easy access to information and accommodation applications
- The system shall allow students to upload and manage the required supporting documents for accommodation applications, including identification documents, proof of registration, NSFAS proof of funding, parent/guardian information documents, and proof of payment.
- System shall allow students to view, add, and update their personal information.
- System shall allow management to approve changes made by students to their personal information.
- System shall allow management to view accommodation listings and occupant details
- System shall provide summaries of students visiting the page and viewing accommodations.
- System shall collect reviews, comments, and areas of improvement from students.
- System shall allow management to track units available, furniture, occupant details, deposits, and breakages
- System shall allow visitors to browse accommodation listings without creating an account
- System shall allow users to reset their passwords if they forget them.
- System shall verify users' email addresses during account registration.
- System shall allow users to filter accommodation listings and available rooms according to relevant criteria, such as price, room type, and availability.
- The web application shall provide a Frequently Asked Questions (FAQ) section containing information about accommodation applications, payments, and accommodation services.

- ❖ The web application shall provide maps and directions to help users locate the accommodation properties.

Non-Functional Requirements:

1. **Usability:** Mobile application must provide a responsive and usable interface on supported mobile screen sizes.
2. **Usability:** Web application must be accessible via standard web browsers
3. **Usability:** System must display clear and understandable error messages guiding the user towards correcting input errors.
4. **Usability:** System should have intuitive user interfaces with consistent navigation and clearly labelled controls, requiring minimal use training.
5. **Security:** System must protect student personal information against unauthorized access.
6. **Security:** System must automatically terminate or expire inactive user sessions after a defined period of inactivity.
7. **Security:** Authentication must separate management and student access levels
8. **Availability:** Supabase database must be accessible to both Flutter and ASP.NET applications
9. **Reliability:** System must maintain data integrity across all platforms
10. **Reliability: System** must back up data regularly
11. **Maintainability:** Source code must be organised and documented

Integration Requirements

How Flutter, ASP.NET and Supabase Will Operate as One Integrated Solution

The proposed solution for Big Brother Student Accommodation will consist of three main components working together seamlessly: a Flutter mobile application for students only, a public ASP.NET website for visitors, students and management and a Supabase backend. These components will operate as one integrated system where all data is shared and synchronised in real-time.

Supabase Backend (Shared Database & Authentication):

Supabase will serve as the central backbone of the system, providing database, authentication, real-time subscriptions, row-level security, and API endpoints for both Flutter and ASP.NET applications to interact with.

Flutter Mobile Application (Students only):

The Flutter mobile application will allow students to view accommodations, check room availability, apply for accommodation, make bookings for room viewing, track application status, upload particulars, submit maintenance requests, and submit reviews. Flutter will use the Supabase Dart client to connect directly to the Supabase backend, with all data retrieved and displayed in real-time**.**

ASP.NET Public Website (Students, Management)

The ASP.NET web application will serve three distinct purposes; each designed for a different group of users. The first is the public section, which is accessible to the general public without requiring them to log in. In this section, unregistered students can browse available accommodation listings, view room details such as facilities and pricing, and learn about the organisation's vision, values, and mission. This section acts as the public face of Big Brother Student Accommodation, helping to attract prospective tenants and address the problem of limited visibility and reach. It also reduces the risk of students falling victim to scammers by providing an official, trustworthy source of information.

The second is the student section, which is accessible only to registered students who have created an account and logged in. In this section, students can register and log in, submit accommodation applications, make bookings, and track the status of their applications. This section is also available through the Flutter mobile app, ensuring that students can access the system from either their phone or a web browser. By giving students, a secure and convenient way to apply and monitor their progress, this section addresses the problem of inefficient application processing and the lack of a digital application platform

The third is the admin section, which is accessible only to management (Mr Ngubane) after logging in with secure credentials. In this section, management can view a dashboard summarising key information, manage student applications, manage accommodation units and room details, verify payment records, view basic reports, and manage all student data. This section replaces the current manual processes, giving management full control over operations from one centralised platform. It addresses the problems of scattered data, manual administration, and the inability to respond to student applications efficiently.

All three sections are connected to the same Supabase backend, which means that data is shared and synchronised in real time. When a student applies through the Student Section, management sees it immediately in the Admin Section. When management approves or rejects the application, the student sees the status update instantly in the Student Section or on the Flutter mobile app. This integrated approach ensures consistency, transparency, and efficiency across the entire system.

Integration Flow:

When a student applies through the Flutter app or the website, the data is sent to Supabase, stored in the database, and a real-time update triggers the ASP.NET dashboard so management can see the new application. When management approves or rejects the application through ASP.NET, Supabase updates the status, and a real-time update triggers Flutter so the student can see the status change.

Integration Requirements

Shared Database - Both Flutter and ASP.NET applications shall connect to the same Supabase database

Data Consistency - Information updated on one platform must be immediately available on the other platform

Authentication - User authentication managed through Supabase Auth, enabling login from both platforms

Shared Schema - Database schema designed to support both management and student functions

API Consistency - Both applications use consistent API endpoints and data formats

Real-time Updates - Supabase real-time subscriptions keep both applications synchronised where appropriate

Row-Level Security - Data access restricted based on user roles

File Storage - Supabase Storage used for uploading documents

Feasibility Findings

TECHNICAL FEASIBILITY

Aspect	Findings
Technology	The required technologies (Flutter, ASP.NET, Supabase) are available and free to use
Integration	Supabase provides client libraries for both Flutter and ASP.NET, making integration straightforward
Group Skills	Team members have basic programming knowledge and are willing to learn the required technologies
Development Tools	All necessary development tools (Visual Studio, Android Studio, GitHub) are available

OPERATIONAL FEASIBILITY

Aspect	Findings
Suitability for Stakeholder	The proposed system addresses Mr Ngubane's core problems: student sourcing, application management, and data centralisation
Suitability for Students	Students will benefit from easy accommodation viewing, online applications, and status tracking
Ease of Use	The system is designed to be user-friendly with minimal training required
Adoption	Mr Ngubane is eager to use the system and has agreed to provide feedback during development
Current Process Improvement	The system will replace manual processes with a digital solution, saving time and reducing errors

ECONOMIC FEASIBILITY

Aspect	Findings
Development Costs	The project uses free and open-source technologies (Flutter, ASP.NET, Supabase free tier)
Hosting Costs	Supabase free tier is sufficient for the prototype; no hosting costs required
Tool Costs	All development tools are free (Visual Studio Community, Android Studio, GitHub)
Stakeholder Costs	No cost to the stakeholder for the prototype development

SCHEDULE FEASIBILITY

Aspect	Findings
Time Available	One academic semester (approximately 12-14 weeks)
Workload	The project is divided into three phases: Planning, Design, Development
Team Size	The group has multiple members, allowing tasks to be divided
Milestones	Clear milestones are set for each phase (Phase 1, Phase 2, Phase 3)
Buffer Time	Some buffer time has been included for unexpected delays
Scope Adjustment	Some features have been simplified or removed to fit the timeline

LEGAL FEASIBILITY

Aspect	Findings
Data Protection	Student data will be protected using Supabase authentication and row-level security
Privacy	Student information will not be shared with unauthorised persons
POPIA Compliance	The system will comply with South African data protection laws
Intellectual Property	All code and documentation will be original work by the team
Stakeholder Consent	Mr Ngubane has consented to the project and the use of his business information

ETHICAL FEASIBILITY

Aspect	Findings
Responsible Use of Information	Student information will only be used for the intended purpose (accommodation management)
Fair Treatment	All students will be treated fairly; the system does not discriminate
Honesty	The team will be honest with the stakeholder about what the system can and cannot do
Transparency	The stakeholder will be informed of all important changes
No Harm	The system is designed to help, not harm, the stakeholder or students

FEASIBILITY CONCLUSION

After conducting a thorough feasibility study across all six areas, the team has concluded that the project is feasible with changes. Technical and operational feasibility support the proposed solution, while economic feasibility is supported by the use of free tools and services. Schedule and legal considerations require the team to simplify or remove complex features. Ethical requirements require responsible handling of student information and transparent communication with the stakeholder.

RISK REGISTER

1.1 Risk Register:

	A	B	C	D	E	F	G	H	I	J	K
1	Risk ID	Risk statement	Category	Likelihood (1-3)	Impact (1-3)	Score	Level	Mitigation	Contingency	Owner	Status
6	R5	too many features (eg. online payments)	Scope	2	3	6	Medium	must,should and could have and confirm core scope with stakeholder early	Remove or postpone lower-priority	Person 3/4	Open
7	R6	Github merge conflicts may occur due to 10 members pushing the same repo	Team	3	2	6	Medium	Create branch per member and person 9 reviews pull requests	Revert to last stable main branch and re-apply changes	Person 9	Open
8	R7	A group member may not complete assigned task on time, delaying dependent activities	Team	2	2	4	Medium	Assign clear individual responsibilities and monitioring progress weekly via the progress tracker	Reassigning urgent tasks to available members and record the concern in the tracker	Person 10	Open
9	R8	Poor communication between group members may cause delays in task handover	Team	3	2	6	Medium	Schedule fixed meeting time (eg. every friday) have daily update on Whatsapp group chat and send reminders 24 hours before deadline	If member does not respond 2 members will be responsible for contact the member directly and record communication concern in progress	Person 10	Open
10	R9	Real student personal information may be accidentally uploaded to Github	Data & Security	1	3		Low	Use only fake test data, check commits before push	Immediately delete file purge history and replace with test data	Person 7	Open
11	R10	Limited testing time before the final demonstration could mean booking or approval bugs go unnoticed	Quality	2	3	6	Medium	Schedule dedicated testing time in the MS Project schedule separate from development time	Focus testing on core must-have function (booking and approval) before the demo	Person 8/9	Open
12											
13											

1.2 Top 3 Highest Risks:

	A	B	C
1	High Risk	Why is it Important ?	What must the Group do now?
2	1. R02 - Double-booking may occur if availability check logic fails	If availability logic fails,two studenst will be booked for the same room and the stakeholder will reject the prototype	Person 3 must test the Supabase date-overlap query early using 20 fake booking and share screenshot in group
3	2. R04 Limited Supabase experience may delay backend integration between Flutter and ASP.NET	If Supabase integration fails, both Flutter mobile app and ASP.NET web app will not receive data and the whole project stops	Person 9 must complete Supabase tutorial in Week 1 and build a small test integration before full development
4	3. R08 poor communicatuion between group members may cause delays in task handover	With 10 members, late replies on Whatsapp cause next member to wait, this delays dependent activities like ASP.NET admin approval depending on Flutter booking. It also affects MS Project Schedule and cause merge conflicts.	Person 10 to schedule weekly progress meetings every Friday, on teams or physically, enforce dailly updates on Whatsapp and send reminders on deadlines 24 hours before deadline
5			
6			

REQUIREMENTS CHANGE LOG

During Phase 1, several adjustments were made to the project requirements based on feasibility analysis and project constraints. The following changes were documented and approved to ensure the project remains realistic and achievable within the available timeframe and resources:

Payment Processing Integration:

The original requirement for full payment-gateway integration was revised to a simplified payment system in which users are provided with the business's banking details and are required to upload proof of payment for verification by management.

. This change was necessary because implementing full payment processing would have introduced significant technical complexity that exceeded the project's constraints

The team determined that implementing a simplified proof-of-payment system would be more appropriate for the prototype scope. This approach allows users to access the business's banking details and upload proof of payment for management to verify, demonstrating the payment process without the complexity of integrating directly with banking systems or payment gateways.

Automated Contract Signing:

The requirement for automated contract signing was removed and declared out of scope. This decision was made because implementing digital contract signing would require additional legal considerations and technical complexity that are not feasible within the academic project timeline. The team and stakeholder agreed that the prototype should focus on core functionality rather than attempting to implement features that would add unnecessary complexity.

Advanced Analytics and Document Verification:

The requirements for advanced analytics and complex document verification were removed and declared out of scope. The team recognised that building complex dashboards with sophisticated data visualisation and document verification would require extensive development effort that could not be justified for a prototype. This change ensures that the development effort is directed towards the most critical features.

Multi-Property Expansion:

The requirement for multi-property expansion was removed and declared out of scope. The team and stakeholder agreed that the prototype should focus on a single property (Big Brother Student Accommodations) to keep the project manageable within the academic timeline.

REFINED PROJECT SCOPE

PROJECT SCOPE INTRODUCTION

Big Brother Student Accommodation is a student housing business located in the Universitas area of Bloemfontein, owned by Mr Ngubane. The business faces significant challenges in sourcing students, managing bookings and payments, processing applications, and ensuring the security of student information. This project aims to develop an integrated software prototype that will digitise and streamline the accommodation management processes, improve student sourcing, enhance communication, and reduce administrative overhead.

The solution will consist of a Flutter mobile application for students and an ASP.NET web application for management, supported by a shared Supabase backend to ensure information consistency across both platforms.

PROJECT AIM

Develop an integrated Flutter mobile application and ASP.NET web application, supported by Supabase, to make accommodation information, applications, bookings, and management easier to handle.

Deliverables:

- Flutter mobile application for the management and students
- ASP.NET web application for management
- Supabase shared backend and database
- Integrated solution where both applications access the same data
- Technical documentation
- User guides where applicable

Acceptance Criteria:

- Management can view and manage student applications
- Management can control, monitor, and manage data
- Management can access, analyse, and respond to student information
- Management can view accommodation listings and occupant details
- Visitors can view accommodation information and room details without creating an account.
- Students can upload particulars?? and apply on the system
- Students can view accommodations, check availability, and make bookings
- Students can view, add, and change their information (with management approval)
- Students can submit reviews, comments, and recommendations
- Students can submit maintenance requests
- Students can view notifications
- Students can upload the necessary supporting documents for their accommodation applications.
- Student information is secured and not accessible by unauthorised persons
- All information is consistent across mobile and web applications
- System is functional for demonstration purposes

Exclusions:

- Full payment processing integration (e.g., automated EFT reconciliation)
- Advanced reporting and analytics dashboards
- Integration with external systems (e.g., banking APIs, municipal systems)
- Full production deployment and hosting
- Professional security audit and penetration testing
- Mobile app store publication
- Multi-property support (limited to Sunrise Student Accommodation)

Project Constraints:

- **Time:** Development must be completed within one academic semester
- **Budget:** Limited budget (student project)
- **Technology:** Stack is predetermined: Flutter, ASP.NET, Supabase
- **Skills:** Team comprises of IT students possessing Flutter, Dart, ASP.NET, Github and Supabase skills necessary for development of the system.

Project Assumptions:

- Stakeholder will be available for requirements gathering, feedback, and evaluation
- Student users will be accessible for feedback and testing
- Supabase will provide sufficient functionality for the project scope
- The approved prototype does not need to be production-ready
- All development work will be managed through GitHub
- Microsoft Project will be used for project scheduling

SUPPORTING EVIDENCE

Method - WhatsApp and direct interview

Changes Discussed - Simplification of payment processing, removal of automated contracts, exclusion of advanced reporting and analytics dashboards for the time being, and scope limitation to one property

Reason for Changes - Technical complexity, legal issues, and time constraints

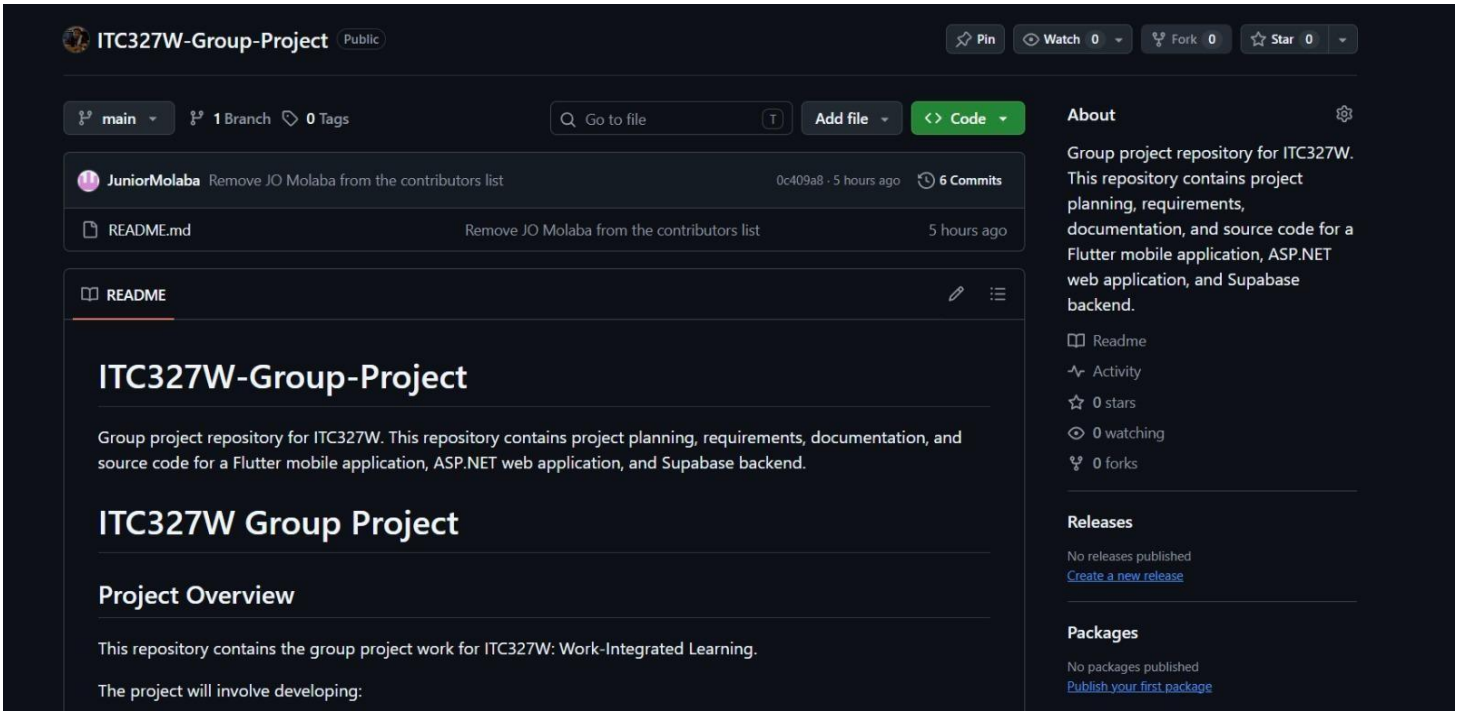
Stakeholder Comments - Mr Ngubane agreed with the changes and confirmed they are acceptable for the prototype

Changes Accepted? – Yes

Evidence Available:

1. GITHUB REPOSITORY EVIDENCE

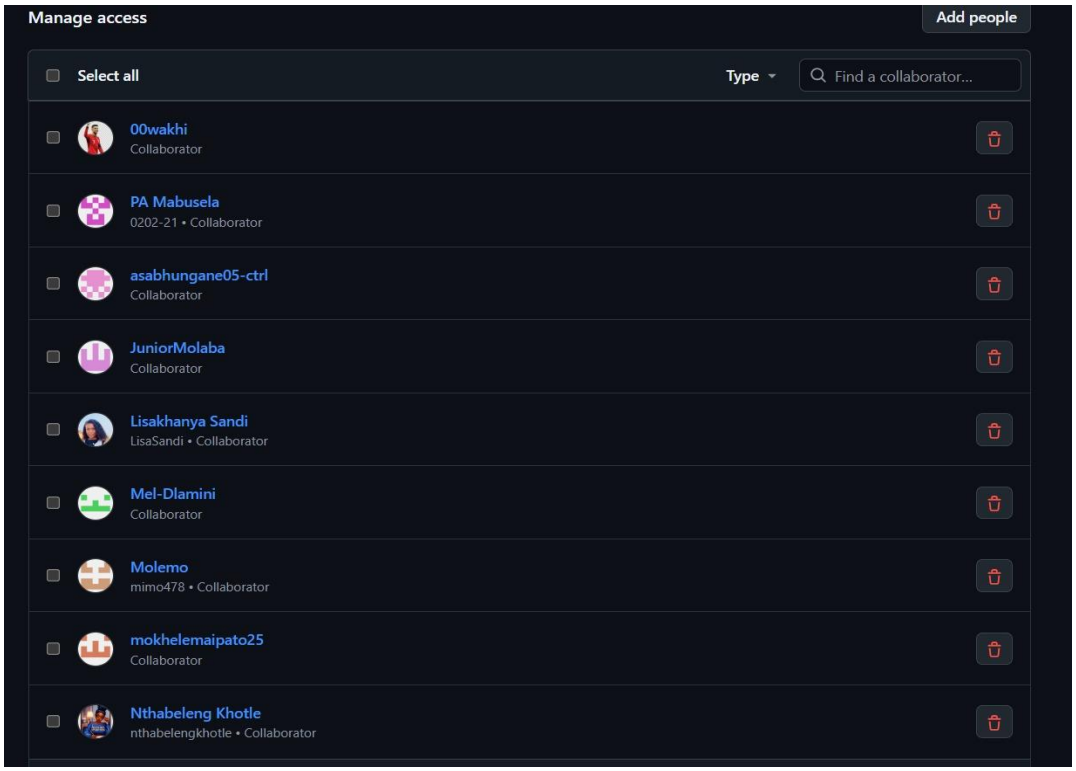
1.1 Repository Home Page:



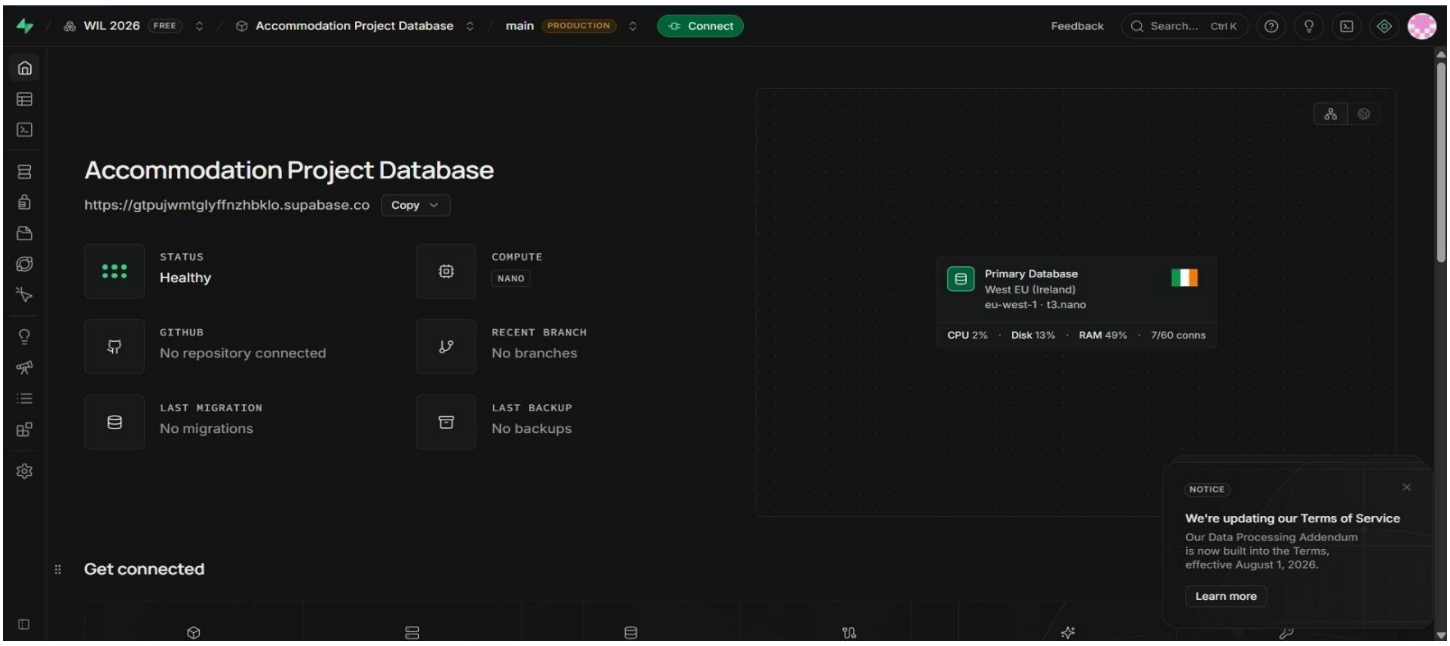
1.2 Repository link:

https://github.com/NhlanhlayethuMazibuko/ITC327W-Group-Project.git

1.3 Team Members/Collaborators:



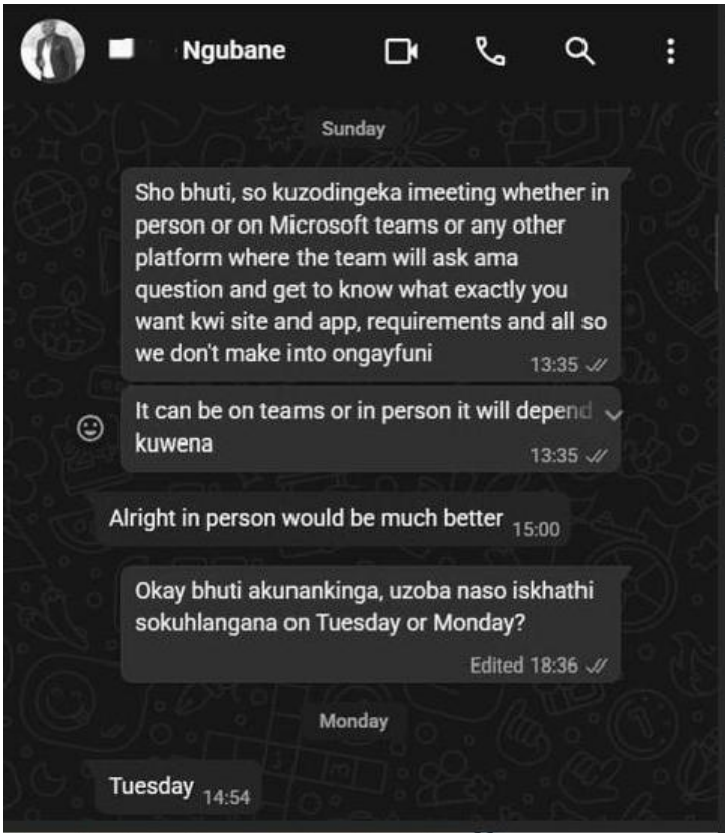
1.4 SUPABASE EVIDENCE



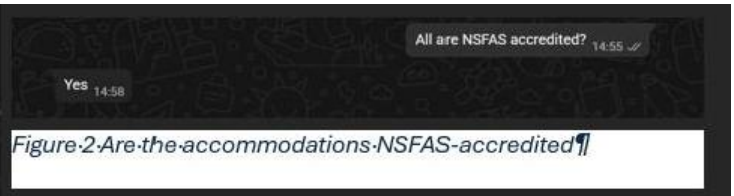
Supabase Link:

https://supabase.com/dashboard/project/gtpujwmtglyffnzhbklo

2. STAKEHOLDER EVIDENCE AND COMMUNICATION



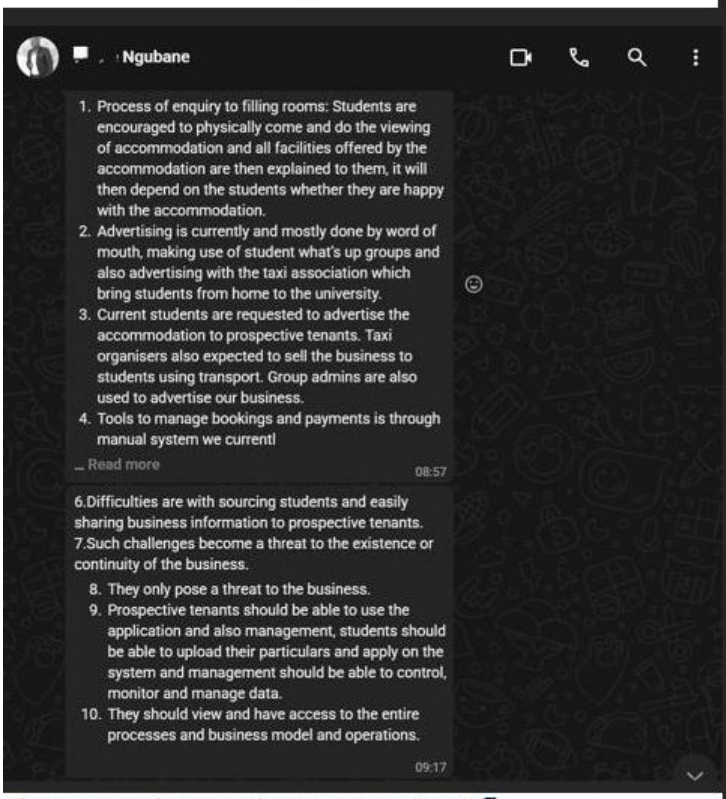
Figure·1·Stakeholder·meeting·confirmation



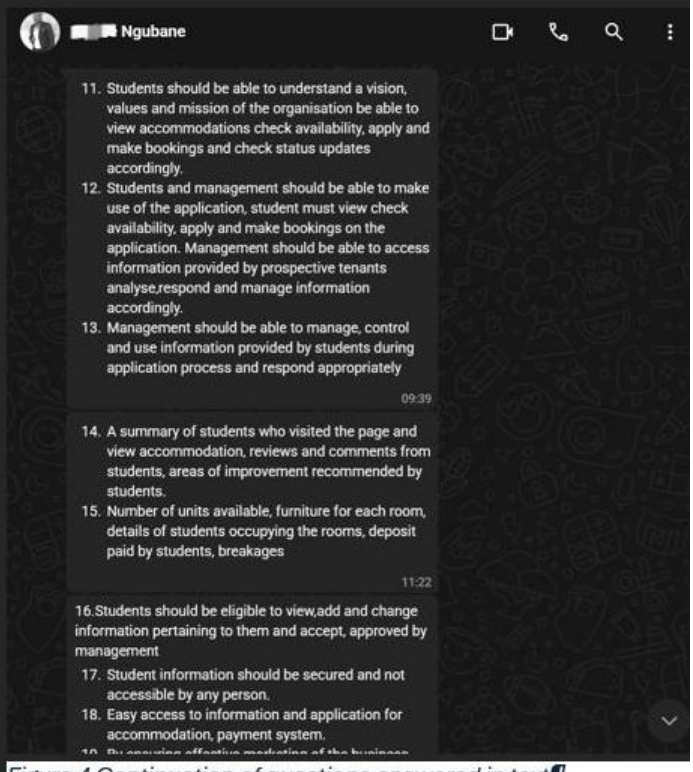
Figure·2·Are·the·accommodations·NSFAS-accredited¶



Figure·3·Time·meeting·started·and·where·it·was·held¶



Figure·5·Interview·questions·answered·in·text¶



Figure·4·Continuation·of·questions·answered·in·text¶

Phase 2

SYSTEM ARCHITECTURE AND DESIGN

System Architecture Design Diagram

This document defines the system architecture of the Big Brother Student Accommodation Management System. It is based on the approved Phase 1 requirements and refined project scope, in which the Flutter mobile application serves students and the ASP.NET web application serves management. It shows how the Flutter application, the ASP.NET application and the shared Supabase backend operate as one integrated system.

System Architecture Diagram

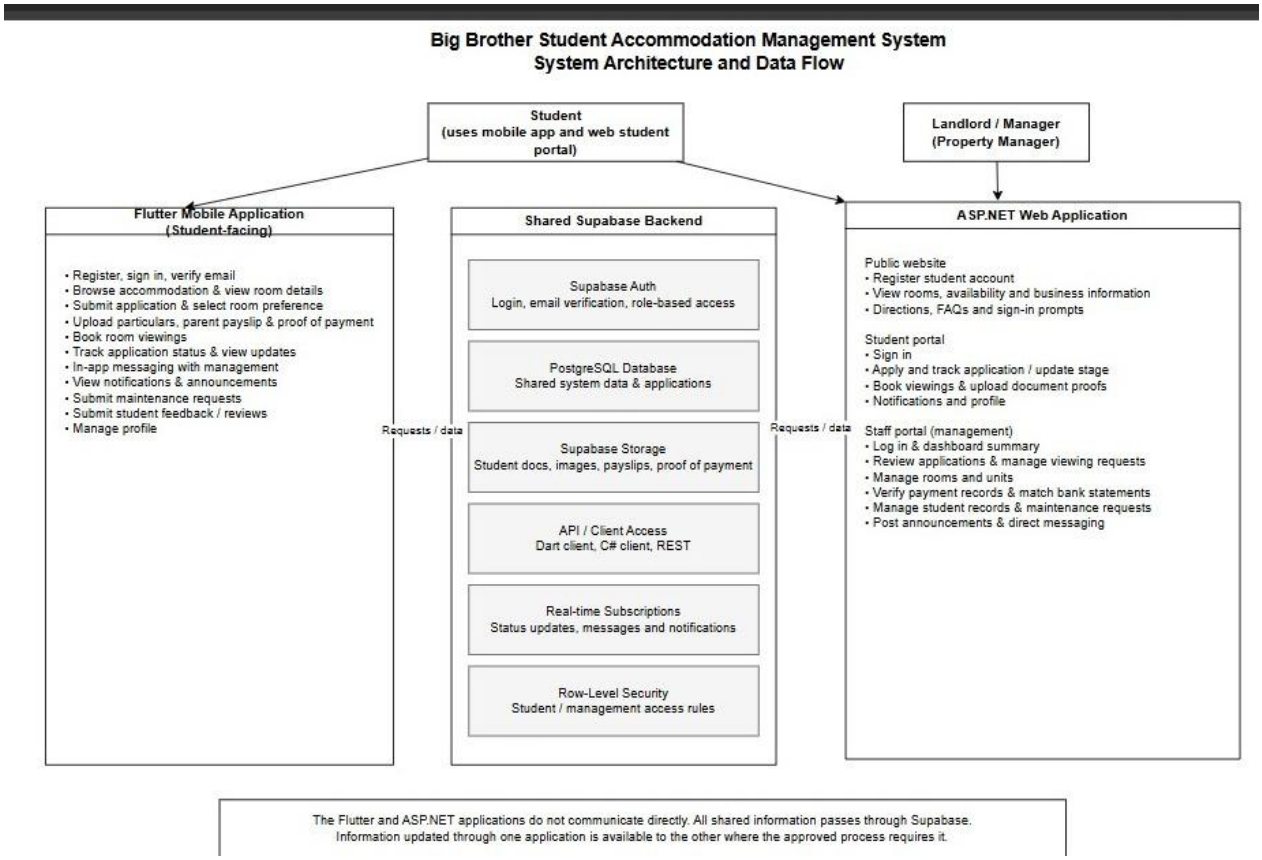


Figure 1: System Architecture Diagram

The diagram shows the two user roles, the Flutter mobile application, the ASP.NET web application and the shared Supabase backend with its services. Requests and data flow from each application to Supabase, and results and updates flow back. The two applications do not communicate directly; all shared information passes through Supabase.

Users and Roles

USER / ROLE	USES FLUTTER?	USES ASP.NET?	MAIN PURPOSE
Student (Unregistered/Registered)	Yes	Yes	Create an account and upload particulars, view accommodation and check availability, apply and make bookings, track application status, view, add and change information (subject to management approval), and submit reviews and feedback.
Landlord/Manager	No	Yes	Control, monitor and manage data; review and respond to applications and student information; approve student changes; manage accommodation listings, units, occupants, deposits and breakages; track simulated payment status.

Components and Responsibilities

COMPONENT	DESCRIPTION	MAIN RESPONSIBILITIES
Flutter Mobile Application	Mobile application used by unregistered and registered students	Account creation and login; upload particulars; view accommodation and availability; apply and make bookings; track application status; view, add and change student information; submit reviews and feedback; view business information.
ASP.NET Web Application	Web application used by management and students	Management authentication; review and approve or reject applications; manage student information and approve student changes; manage accommodation listings, units and occupants; track deposits, breakages and simulated payment status; view accommodation viewing summaries. Students will be able to create accounts and login; upload particulars; view accommodation and availability; apply and make bookings; track application status; view, add and change student information; submit reviews and feedback; view business information.
Supabase Authentication	Central authentication service.	Verifies users and separates student and management access.
Supabase Database	Shared PostgreSQL database.	Stores all system records used by both applications.
Supabase Storage	Shared file storage.	Stores files uploaded by students, such as their particulars and supporting documents.
Supabase API / Real-time	Access layer and update notifications.	Lets both applications read and write shared data and receive updates when data changes.
Supabase Row-Level Security	Data access rules.	Restricts each user to the records they are allowed to access, so students see only their own information.

Main Data Flows

FROM	TO	INFORMATION EXCHANGED	REQUIREMENTS
Student via Flutter	Supabase Authentication	Registration and login details.	FR1, FR10
FROM	**TO**	**INFORMATION EXCHANGED**	**REQUIREMENTS**
Supabase Authentication	Flutter / ASP.NET	Verified session and user role (student or management).	FR10, NFR5
Student via Flutter	Supabase Storage	Uploaded particulars and supporting documents.	FR1
Student via Flutter	Supabase Database	Student details, applications, bookings, information changes, reviews and feedback.	FR1, FR2, FR4, FR14, FR18
Supabase Database	Flutter	Accommodation listings and availability, application status, approved student information and business information.	FR3, FR4, FR5, FR8, FR9, FR12
Management via ASP.NET	Supabase Authentication	Management login credentials.	FR10
Supabase Database / Storage	ASP.NET	Applications, student information and documents, bookings, units, occupants, deposits, furniture, breakages and viewing summaries.	FR6, FR7, FR16, FR17, FR19
ASP.NET	Supabase Database	Application decisions, approval of student changes, unit and occupant updates and simulated payment status.	FR2, FR6, FR13, FR15, FR19
Supabase Real-time	Flutter / ASP.NET	Notification that shared data has changed.	FR5, Integration Requirements

How the Components Operate as One System

Flutter users: Students use the Flutter mobile application.

ASP.NET users: Management (owner or property manager) and students uses the ASP.NET web application.

Supabase: Supabase is the shared backend. It verifies users, stores all records in one PostgreSQL database, stores uploaded files, restricts access with Row-Level Security, and sends real-time updates to both applications.

Flutter and Supabase: Flutter connects to Supabase through the Supabase Dart client.

ASP.NET and Supabase: ASP.NET connects to Supabase through the Supabase C# client or REST API and uses the same database tables as Flutter.

One integrated solution: The two applications never communicate directly. Because both use the same Supabase project, a record created in one is available in the other. For example, when a student applies in Flutter, the application is stored in Supabase and management sees it in ASP.NET. When management approves or rejects it, Supabase updates the status and Flutter shows the decision to the student. Payments are simulated payment-status data only, with no external banking or payment gateway.

Use Case Diagram

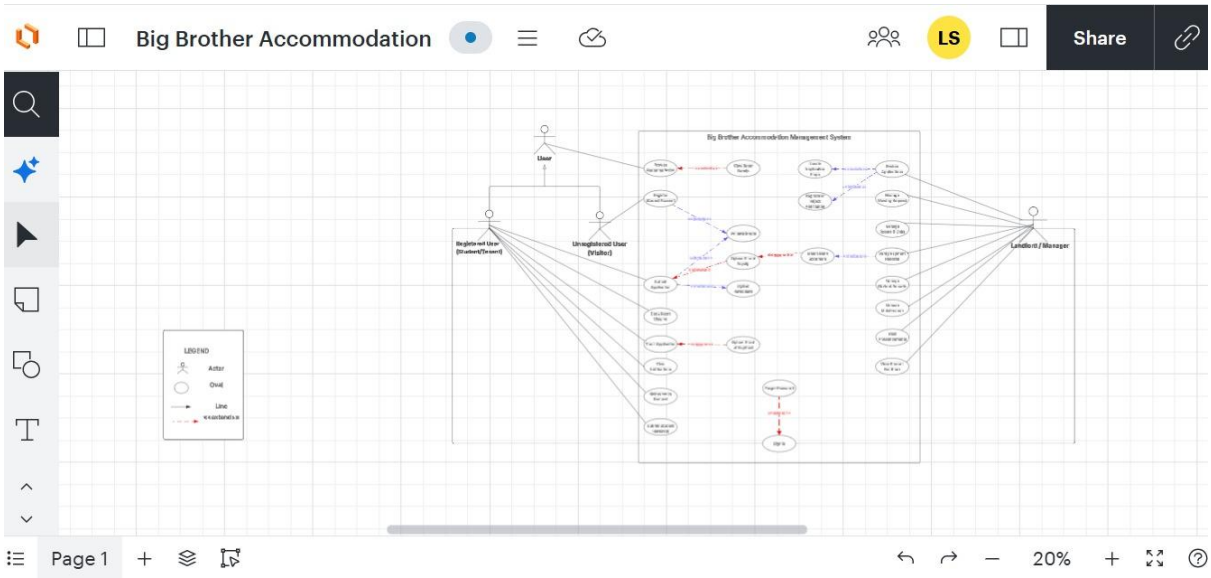


Figure 2: UML Use Case Diagram

The use case diagram illustrates how Students/Tenants and Landlords/Managers interact with the Big Brother Accommodation Management System. Students can search and apply for rooms, make payments, upload proof of payment, and track applications, while landlords manage room listings, review applications, verify payments, assign rooms, and communicate with tenants.

Sequence Diagrams

Mobile Application

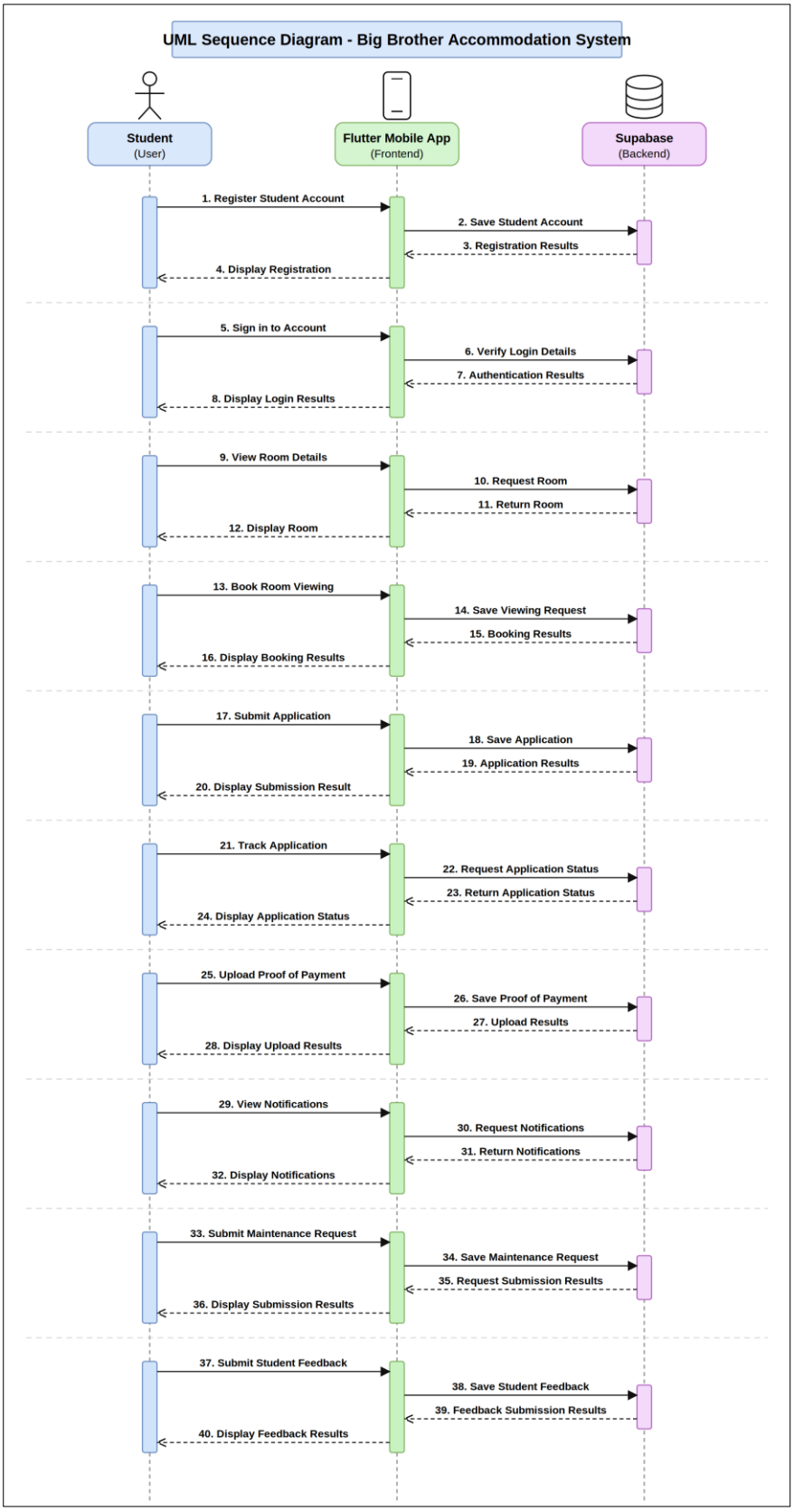


Figure 3: Mobile Application Sequence Diagram

The Student UML Sequence Diagram shows how a student interacts with the accommodation system. The student uses the app to register an account, sign in, view room details, book a room viewing, submit an application, track its progress, upload proof of payment, view notifications, submit maintenance requests, and provide feedback.

For every interaction, the student sends a request to the app, the app communicates with the database, and the database returns the result through the app to the student.

Management is not included in this diagram because it focuses only on the student. Management creates the notifications, while the student only views them.

Web Application

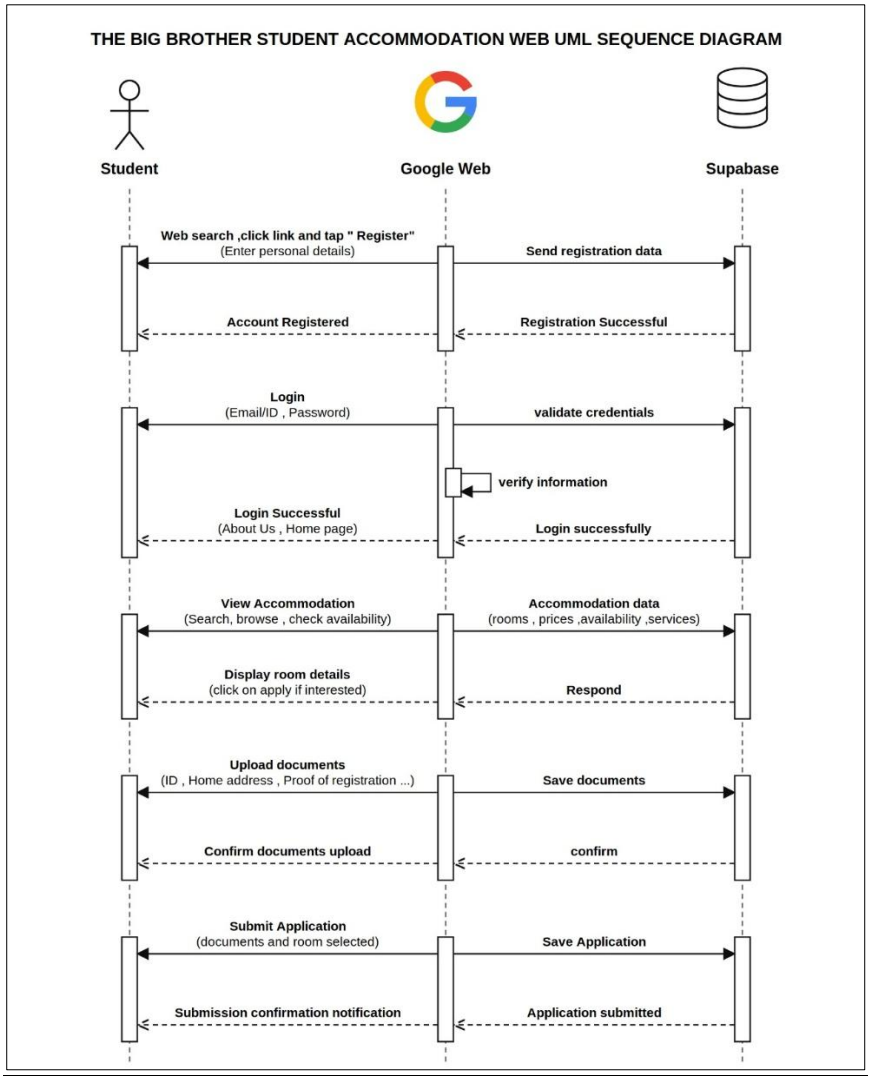


Figure 4: Web Application Sequence Diagram

This Web Sequence Diagram illustrates how every student in demand of an accommodation will be able to access information about rooms availability, services rendered and costs digitally, on google. Track their applications as well which will be approved or rejected by the landlord.

Entity Relationship Diagram

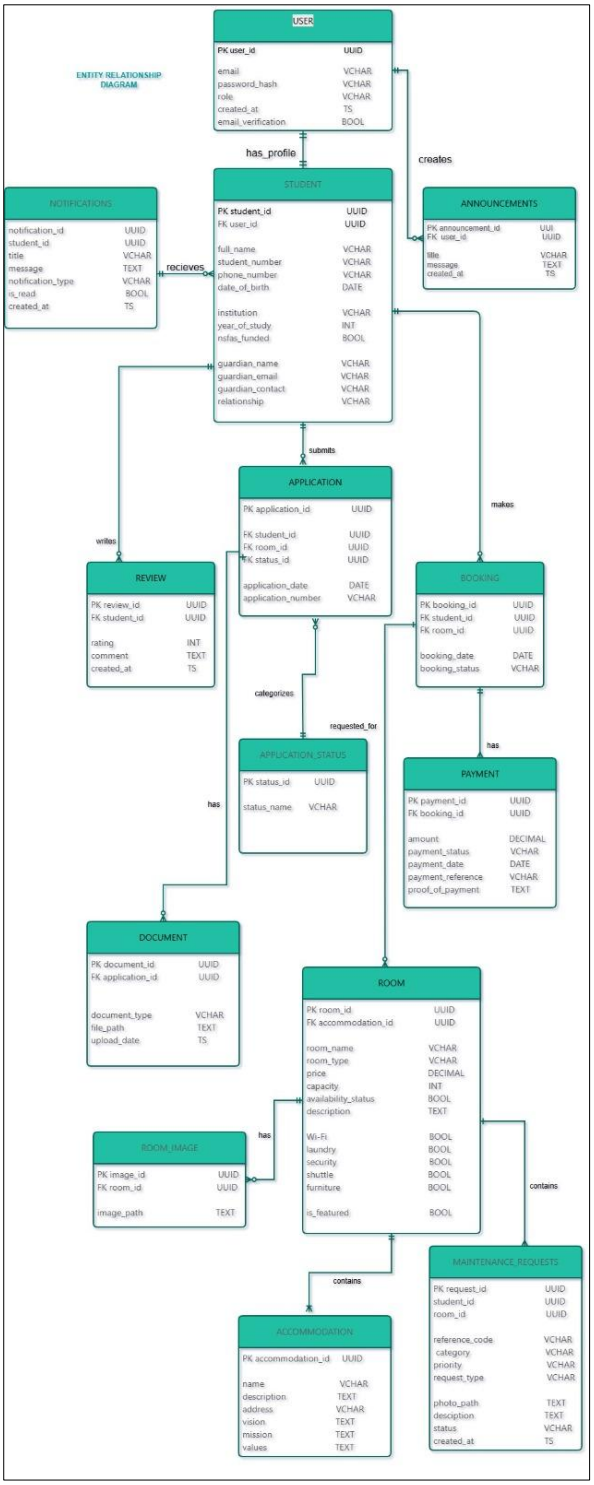


Figure 5:Entity Relationship Diagram

The Big Brother Student Accommodation Management System uses a shared Supabase relational database to support both the Flutter mobile application and ASP.NET web application. The database stores user accounts, student information, accommodation details, applications, bookings, payments, maintenance requests, notifications, announcements, and supporting documents.

Entity Summary:

Entity	Purpose
USER	Stores authentication and account information
STUDENT	Stores personal, academic, and guardian information
ACCOMMODATION	Stores accommodation details and information
ROOM	Stores room information, pricing, facilities, and availability
APPLICATION	Stores accommodation applications submitted by students
APPLICATION_STATUS	Tracks application statuses.
DOCUMENT	Stores uploaded application documents.
BOOKING	Stores room booking information
PAYMENT	Stores booking payment records.
REVIEW	Stores student feedback and ratings
MAINTENANCE_REQUEST	Stores maintenance requests submitted by students.
NOTIFICATION	Stores system notifications sent to students
ANNOUNCEMENT	Stores accommodation announcements
ROOM_IMAGE	Stores room images displayed in the system

Business Rules

A student must have a registered user account before submitting an application. A room must belong to accommodation. An application must be linked to one student and one room. A booking must reference a valid room. A payment must be linked to a booking. A maintenance request must be linked to a student and a room. Notifications are linked to individual students. Room images cannot exist without a room.

Normalization

The database design follows normalization principles to minimize redundancy and improve consistency. Separate entities such as APPLICATION, BOOKING, DOCUMENT, PAYMENT, MAINTENANCE_REQUEST, NOTIFICATION, and ROOM_IMAGE ensure efficient data management and referential integrity.

User Interface Designs

Mobile User Interface Design

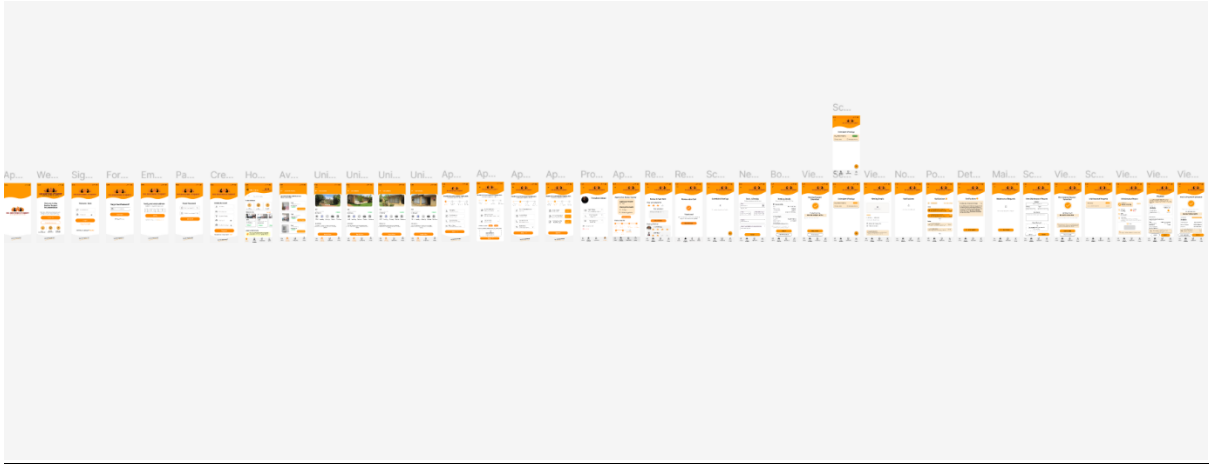


Figure 6:Flutter Mobile Screens

Web User Interface Design



Figure 7.1: Web Admin Portal User Interface

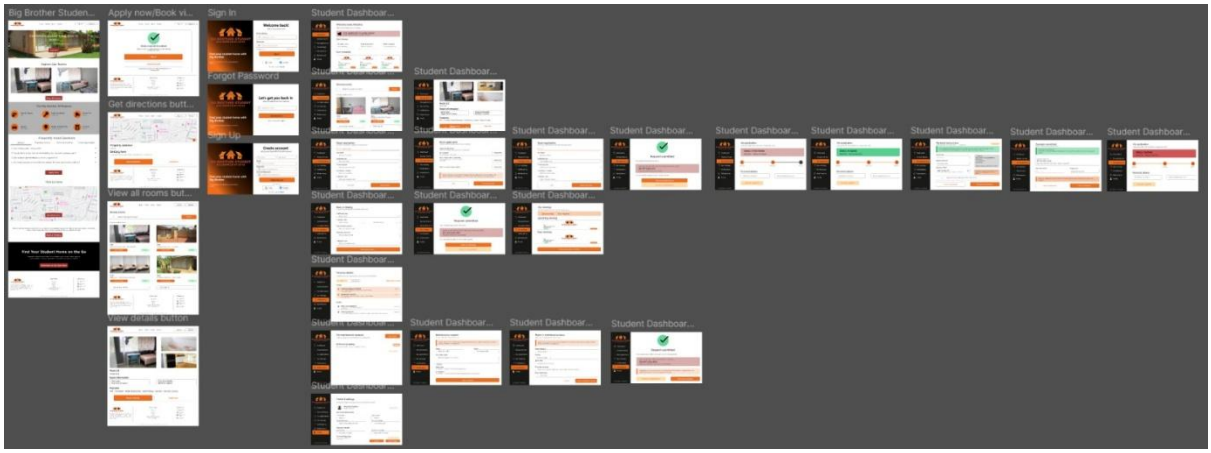


Figure 7.2:Web Student Portal User Interface

DESIGN ALIGMNMENT TABLE

Requirement ID	Functional Requirement	Use Case / Sequence	ERD Entities	Interface Screen
FR1 / FR10	Register & Verify Account	Register Student Account (Seq. 1–4)	User, Student	Register / Verify Account
FR10	Login & Authenticate	Sign in to Account (Seq. 5–8)	User	Login Screen
FR3 / FR12	View Accommodation Listings	View Room Details (Seq. 9–12)	Accommodation, Room, Room Image	Home / Room Listings / Room Details
FR4	Book Room Viewings	Book Room Viewing (Seq. 13–16)	Booking, Room	Book Viewing Screen
FR2	Submit Application	Submit Application (Seq. 17–20)	Application, Application Status	Application Form (4 Steps)
FR5	Track Application Status	Track Application (Seq. 21–24)	Application, Application Status	Application Status Tracker
FR13	Upload Proof of Payment	Upload Proof of Payment (Seq. 25–28)	Payment, Document	Payment / Proof Upload
FR5	View Notifications	View Notifications (Seq. 29–32)	Notifications	Notifications Screen
FR18	Submit Maintenance Request	Submit Maintenance Request (Seq. 33–36)	Maintenance Requests	Maintenance Request Form
FR18	Submit Student Feedback	Submit Student Feedback (Seq. 37–40)	Review	Feedback / Review Screen
FR6 / FR7 / FR16	Management Data Control & Review	Admin Review & Approval Flow	Application, Room, Student, Booking	ASP.NET Staff Dashboard
FR19	Track Units, Furniture & Breakages	Admin Property Management	Room, Accommodation	ASP.NET Properties & Rooms

Notes

- Interface names are shortened for readability and are based on the provided web portal and mobile app screens.
- Staff functions are represented by the ASP.NET staff dashboard; student-facing functions are represented by the mobile app screens.
- A dedicated furniture and breakage inventory screen was not clearly shown in the provided screens.

DIAGRAM DESIGNS TABLE

Design	Tool Used	Source Link
System Architecture Design Diagram	draw.io	https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&target=blank&highlight=0000ff&edit=_blank&layers=1&nav=1&title=System%20Architecture%20Design%20Diagram.drawio&dark=0#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1WDEojn6uXPUyXhGfElotnWxwkxLsXJ7z%26export%3Ddownload
Use Case Diagram	Lucid chart	https://lucid.app/lucidchart/7283e1a9-e931-4707-8e40-278ad39ba254/edit?viewport_loc=-1701%2C-621%2C4230%2C1712%2C0_0&invitationId=inv_4277887e-2340-4d31-94b6-65d9a399f9d7
Sequence Diagram (Mobile)	draw.io	https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Untitled%20Diagram.drawio&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1pcUpu_bFWETnkdJYSbbd4mf9SvOqUl1t%26export%3Ddownload
Sequence Diagram (Web Application)	draw.io	https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&target=blank&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Sequence%20Diagram(APS.NET).drawio&dark=0#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1pe3ZnGeT62MtFIwFBITfMiQIEaAiwgCh%26export%3Ddownload
Entity Relationship Diagram	draw.io	https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Entity%20Relationship%20Diagram.drawio&dark=0#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1hPSEFW3vkxjzCYKUqcsKeAocBAP2HUcu%26export%3Ddownload
Flutter wireframes	Figma	https://www.figma.com/design/wn4ll8WEIqOOEAMzvopEw1/Big-Brother-Student-accommodation-App?node-id=0-1&t=rUtyROwJk860PPph-1
ASP.Net wireframes	Figma	https://www.figma.com/design/nsfe2RnSzAZCK1aHrkBLQn/Big-Brother-Student-Accom.?node-id=2479-2&t=xBQDfkAT5m9ywMLk-1

MICROSOFT PROJECT PLAN

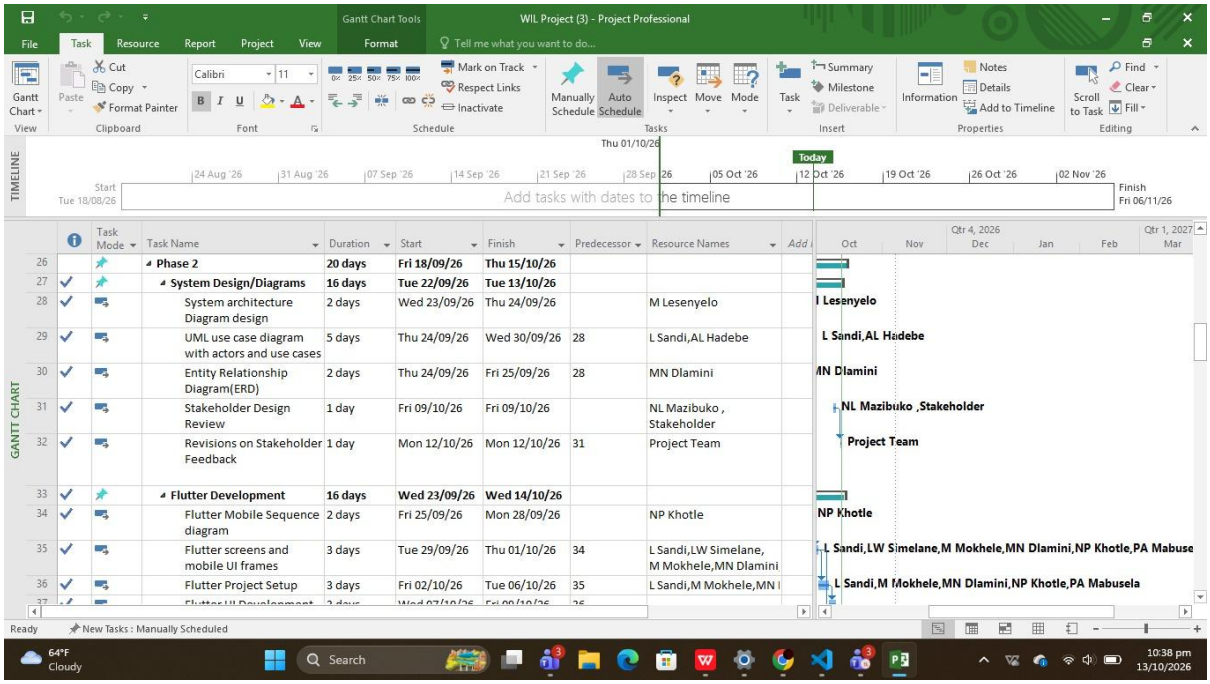


Figure 8:Microsoft Project Screenshot

Github Link

https://github.com/NhlanhlayethuMazibuko/ITC327W-Group-Project

Stakeholder Evidence and Communication

